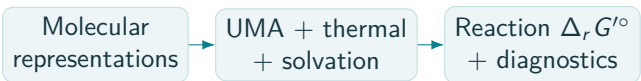
Metabolic reaction thermodynamics

MetaG

From molecular structure to metabolic reaction Gibbs energy

Pipeline and benchmark performance



Ray Zhu

Collaborator discussion 7 October 2026

Scope and thermodynamic convention

Aim

Use UMA as the electronic-energy core, with physically justified molecular states and thermodynamic corrections.

Apply the same chemical rules to unseen ModelSEED reactions. Every result should explain its representation, energy terms, and limitations.

Target conditions

- $T = 298.15$ K; pH 7
- Ionic strength $I = 0$
- No Mg-binding contribution
- Solutes at 1 M; pure liquid water

For metabolite concentrations, a separate reaction-quotient term is needed: $\Delta_r G' = \Delta_r G'^{\circ} + RT \ln Q'$.

Evidence: README; pipeline.py (CONDITIONS); assembly.py. Conditions are fixed in the current code.

Where empirical information enters

Electronic model

UMA is trained on quantum-chemical data (OMol25). It predicts energies and forces for specified geometries, charge, and spin.

Thermodynamic model

Conformers, thermal corrections, solvation, chemical states, standard states, and stoichiometry convert those energies into $\Delta_r G'^{\circ}$.

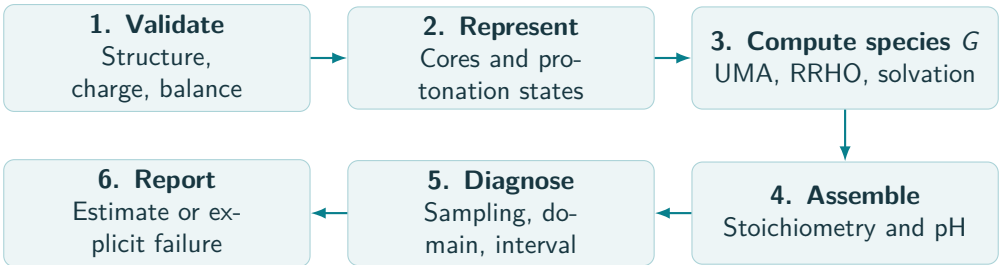
Empirical constants

pKa references, the experimental hydration free energy of water, and carbonyl hydration constants. openTECR is used for evaluation and interval calibration.

No openTECR reaction-energy offset is fitted in the point estimate. Routing rules were developed with openTECR in view, so openTECR scores are development results, not independent validation.

Evidence: production_policy.py; pka_transform.py; aldehyde_hydration.py; config.py (ANCHOR_CORRECT off).

Pipeline



Experimental labels enter only evaluation and uncertainty calibration, after point-energy assembly.

Evidence: routing/planner.py; energetics/; assembly.py; pipeline.py; uncertainty.py.

Exact composition of the reaction energy

$$\Delta_r G'^{\circ} = \sum_i \nu_i \, G_i^{\mathrm{eff}} \;+\; n_{\mathrm{H^+}} \, G_{\mathrm{H^+}} \;+\; \sum_{j \in \mathrm{sites}} s_j \, m_j \, \Delta G_j^{\mathrm{pKa}}$$

Solute (core or full)	$G_i = -RT \ln \sum_k e^{-(E_k^{\mathrm{UMA}} + G_k^{\mathrm{RRHO}} + \Delta G_k^{\mathrm{ALPB}})/RT} \;+\; RT \ln 24.46$
Carbonyl hydration	$G_i^{\mathrm{eff}} = -RT \ln \left[e^{-G_i/RT} + \sum_S e^{-(G_S - n_S G_{\mathrm{w}})/RT} \right]$
Liquid water	$G_{\mathrm{w}} = E^{\mathrm{UMA}} + G^{\mathrm{RRHO}} + \Delta G_{\mathrm{hyd}}^{\mathrm{exp}} + RT \ln 55.34 + RT \ln 24.46$
Proton at pH 7	$G_{\mathrm{H^+}} = -26.3 - 1104.5 - RT \ln 10 \cdot \mathrm{pH} = -1170.8$
pKa site j	$\Delta G_j^{\mathrm{pKa}} = -RT \ln 10 \; \log_{10} \left(1 + 10^{\pm(\mathrm{pH} - pK_{a,j})}\right)$

k: validated conformer minima. S: distinct hydrates with n_S waters. $s_j = \pm 1$ product/reactant, m_j multiplicity, + acid / − base. $RT \ln 24.46 = 7.93$ (1 atm→1 M); $RT \ln 55.34 = 9.96$; $\Delta G_{\mathrm{hyd}}^{\mathrm{exp}} = -26.4$ kJ/mol. G^{RRHO}: ZPE + ideal-gas terms at 1 atm, exact UMA Hessian, 50 cm^{-1} floor, detected linearity and symmetry number. **Off by default:** class anchors, SMD correction, hydro-lyase water patch, ring–chain forms, qRRHO.

Evidence: assembly.assemble_reaction; pipeline.py (G_HPLUS, STD_STATE_KJ, water_ref_G); protonation.site_free_energy; aldehyde_hydration.mixture_G_states; config.py.

Conformers and thermal ensemble

Sampling

- ETKDG/MMFF pool, ranked by UMA single points
- Lowest candidates relaxed with UMA; ALPB at the same geometry
- Batches added until the ensemble G plateaus
- Independent seed sequences available for convergence checks

Minimum validation

- Every sampled candidate must be a validated minimum
- Imaginary modes: retighten, then follow the mode both ways
- Retry with 1200 optimization steps if unresolved
- Still unresolved, or > 64 candidates: species fails explicitly

The within-run spread U_{samp} does not bound unsampled basins; flexible molecules need independent-seed checks.

Evidence: energetics/sampling.py; energetics/thermal_ensemble.py; pipeline.py (THERMAL_VALIDATION_MAX, THERMAL_RETRY_STEPS).

Checks on every species

Identity

- Connectivity and specified stereochemistry verified after every relaxation
- Explicit-water clusters: solute identity and intact waters
- Solvation, RRHO and electronic energy from the same geometry

Failure reporting

- Unsupported chemistry is refused, not approximated
- No-estimates state the cause: time budget, unresolved minimum, energy failure, quarantined pKa
- Charge-closure and $|\Delta_r G'^{\circ}|$ sanity guards fail closed

A result with $\Delta_r G'^{\circ}$ = null is a failed estimate, not zero energy.

Evidence: energetics/conformers.py (same_stereochemistry); pipeline.score_reaction guards; _no_estimate.

Structural cores: CoA example

Current CoA-core representations preserve the thioester

R: identical CoA-derived spectator (charge −4). Teal: preserved thioester.

a rxn02168 · (R)-C4

Input (spectator abbreviated)



Routed core (scaffold shown explicitly)



The CoA route retains the thioester and an amide-containing scaffold. These structures come directly from the router.

Evidence: rxn02168 route; balance and spectator identity checked by make_truncation_figure.py.

Truncation acceptance

Structural guards

Element and charge balance
Consistent removed fragments
Intact rings and reactive bonds
Preserved proton accounting

Energetic check

Radius R and $R+1$ must be different molecular models
$|\Delta G_{R+1} - \Delta G_R| \leq$ tolerance, and
$2\sqrt{U_R^2 + U_{R+1}^2} \leq$ tolerance
Otherwise: full molecule

This is a guard, not a convergence proof. Remaining cofactor/NTP core approximations are reported with each result.

Evidence: pipeline._score_trunc_validated; routing/truncate.py; routing/cofactor_truncate.py.

Protonation and chemical states

Protonation

- Reference microspecies computed in QM; pKa terms bring them to pH 7
- Independent-site treatment of multiple sites
- Missing pKa entries are not fabricated
- Unsupported phosphorus environments refused; P–N ladder quarantined

Solvation and hydration

- Gas-phase UMA minima + ALPB solvation
- Liquid water from experiment
- Carbonyl hydrates as distinct species, constants from independent K_{hyd} data
- Ring–chain and explicit-water paths: optional, off

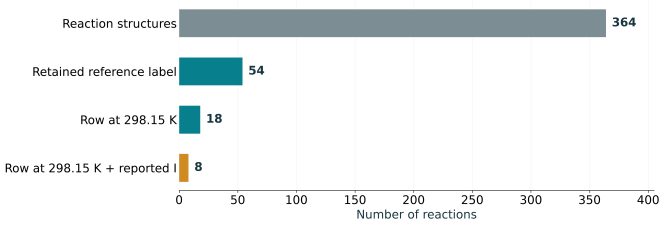
Coupled multi-site protonation needs a map from the QM microstate to measured macrostates; a sequential-pKa routine exists but is not routed.

Evidence: energetics/protonation.py; pka_transform.py; pka_domain.py; aqueous_states.py; docs/pka-provenance.md.

Performance

Accuracy, coverage and baselines

Benchmark panel and reference labels



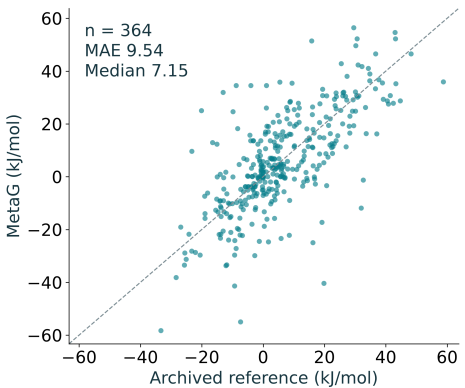
364 openTECR reactions with structures; all are scored.

54 keep a label after strict curation; **8** have a measurement at 298.15 K with reported ionic strength.

The full-panel numbers that follow use archived labels, whose conditions are not all resolved.

Evidence: curated/reference_report.json; src/metag/data/reactions_opentecr_std.json.

Accuracy on openTECR



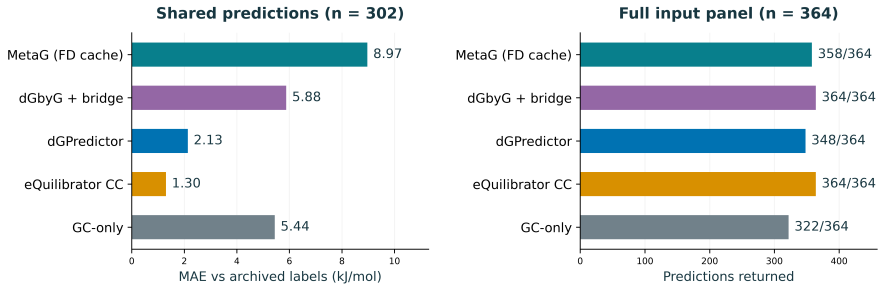
Reactions scored	364 / 364
MAE	9.54 kJ/mol
Median absolute error	7.15 kJ/mol
Errors $>$ 20 kJ/mol	35
Labels inside 95% interval	97%
Median interval half-width	29.7 kJ/mol

Release 2026-10-01c, archived labels. Development result: rules were chosen with this panel in view.

The current version (07e) adds stricter validation and has not yet been re-benchmarked on the full panel.

Evidence: artifacts/results/metag_opentecr_calibrated (364 records; interval coverage in-sample).

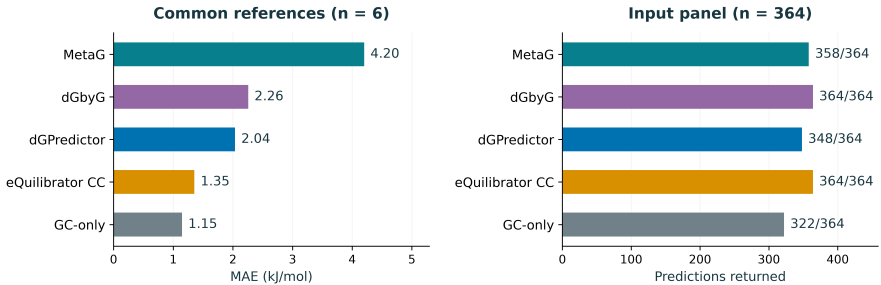
Comparison with pretrained baselines



302 reactions predicted by every method; archived openTECR labels. MetaG withholds 6 phosphorus cases (4 P–N, 2 phosphonate). The baselines are trained on TECRDB data: 148/302 shared reactions match dGbyG training reactions. This is not held-out accuracy.

Evidence: artifacts/benchmark/comparison; Fan et al., Cell Systems (2025), DOI: 10.1016/j.cels.2025.101393.

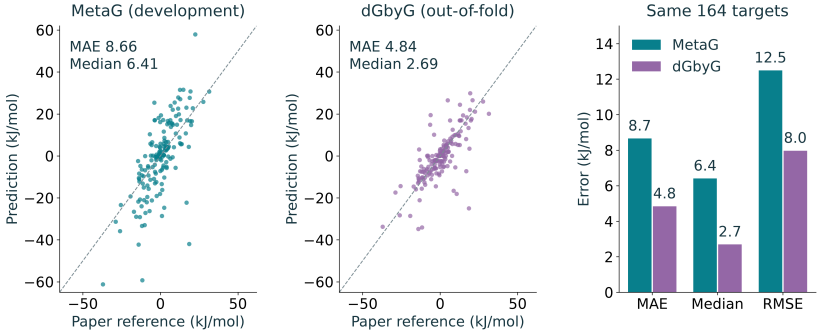
Strict reference conditions (small subset)



Eight reactions have a measurement at 298.15 K with reported ionic strength; six are supported by all methods. All six match dGbyG training reactions.

Evidence: artifacts/benchmark/comparison; native dGbyG I=0.25 M output excluded from the I=0 comparison.

MetaG vs held-out dGbyG on matched reactions

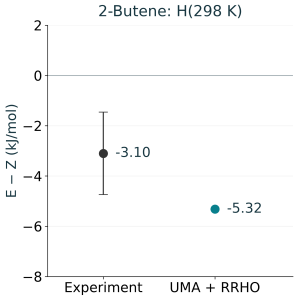


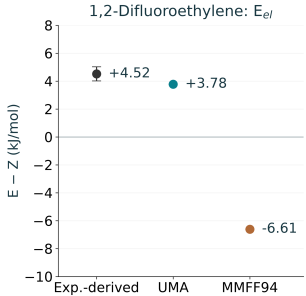
164 dGbyG-paper targets matched by structure, direction and scale. MetaG bridged from $I = 0$ to 0.25 M with eQuilibrator dissociation data.

Held-out dGbyG is more accurate than MetaG here. Only one EC1 reaction is matched.

Evidence: artifacts/benchmark/paper_reproduction/metag_vs_dgbyg_pairs.tsv, metag_pairing_audit.tsv. kJ/mol.

What geometry adds: stereoisomers





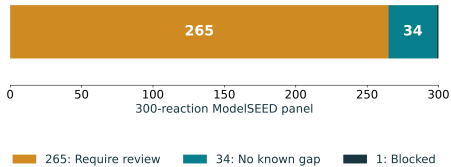
The published dGbyG encoder gives **identical inputs** within E/Z and diastereomer pairs (fumarate/maleate; tartrate forms).

UMA reproduces the opposite orderings of these two gas-phase pairs; MMFF94 gets difluoroethylene wrong.

This limits one published encoding, not graph models in general, and is a gas-phase test, not aqueous reaction accuracy.

Evidence: state_qc_20261007/stereo_reference_verified.json; robustness_20261007/geometry_probe.tsv. NIST/Prosen 1951; Craig et al. 1971, Table II.

ModelSEED coverage



Within 588 unique input structures
238 have multiple acid sites under an independent-site treatment.

117 contain aromatic N requiring speciation review.

Interpretation
CPU preflight flags, not counts of wrong predictions. “No known gap” is not a guarantee of accuracy. 300-reaction panel, not the full database.

Evidence: analysis/code_review/modelseed_preflight.json; python -m metag.tools.preflight.

Summary and limitations

What MetaG provides

- Reaction $\Delta_r G'^{\circ}$ from UMA energies with no fitted reaction offsets
- Every term traceable per species and per reaction
- Explicit failures instead of silent approximations
- Geometry- and stereo-resolved species energies

Limitations

- openTECR MAE 9.54 kJ/mol is a development score
- Held-out dGbyG is more accurate on matched reactions
- Independent-site pKa treatment; P–N chemistry withheld
- Continuum solvation; sampling can miss basins
- Version 07e awaits a full benchmark and recalibration

Evidence: Sources as cited on previous slides.

Appendix: reproducing the dGbyG paper

	MAE	Median	RMSE
dGbyG (10-fold)	9.18	4.11	25.88
eQuilibrator* (LOO)	19.46	5.33	82.88
dGPredictor (LOO)	16.26	5.48	47.85

Reconstructed from archived out-of-fold predictions; matches the published headline metrics and all 18 EC-class MAEs.

* Paper's regression protocol, not today's component-contribution API.

Protocol details

- Headline uses a 675-target historical set, not the 671-target release
- dGbyG: shuffled 10-fold; baselines: LOOCV on 679 targets
- Ridge α chosen globally, not nested
- dGbyG does not win every EC class: EC1 48.7 vs eQuilibrator 38.1

Evidence: tools/thermo_benchmark/paper_reproduction/REPRODUCTION.txt; artifacts/benchmark/paper_reproduction/. kJ/mol.

Appendix: reference transformation

$$\Delta_r G'^{\circ}(c_\star; T) = s[-RT \ln K'_{\mathrm{obs}}] + \mathcal{L}_r(c_\star; T) - \mathcal{L}_r(c_{\mathrm{obs}}; T)$$

- s aligns the experimental reaction orientation with ModelSEED.
- $c_\star$: pH 7, $I = 0$, pMg 14 (approximately negligible free Mg).
- $\mathcal{L}_r$ uses dissociation/binding and ionic-strength information; no fitted formation energy enters.
- Temperature remains the measured temperature. Missing I is imputed at 0.25 M and flagged.

Evidence: validation/reference_transform.py; tools/build_reference.py; eQuilibrator 3.0.

Appendix: reference curation

- Exclude unknown free Mg activity; never substitute total Mg for free Mg
- Exclude unresolved constant annotations
- Record source rows and corrections for every retained label
- 268 retained rows have multiple matching candidates; ambiguity is not proof of a wrong assignment

Evidence: build_reference.py; reference_transform.py; curated/reference_report.json.

Appendix: protonation partition functions

For two independent acidic sites relative to a fully protonated reference:

$$Z_{\text{independent}} = (1 + 10^{\text{pH} - pK_{a,1}})(1 + 10^{\text{pH} - pK_{a,2}}).$$

For two sequential *macroscopic* dissociation constants:

$$Z_{\text{sequential}} = 1 + 10^{\text{pH} - pK_{a,1}} + 10^{2\text{pH} - pK_{a,1} - pK_{a,2}}.$$

The independent product contains an additional term. Production uses the independent form; the sequential form is implemented but not routed.

Evidence: energetics/protonation.py; eQuilibrator 3.0, DOI: 10.1093/nar/gkab1106; Ullmann, DOI: 10.1021/jp026454v.

Appendix: code and tests

Structure

- CPU-only routing separated from QM
- Pure reaction-energy assembly
- Shared sampling and thermal bookkeeping
- Configuration fingerprint in every result and cache key

Not yet in place

- Immutable per-job configuration
- Dependency identities in cache provenance
- Reliable batch resume and cost accounting

387 tests passed, 1 opt-in GPU test skipped (version 07e). This verifies implementation behavior, not predictive accuracy.

Evidence: state_qc_20261007/tests_final.log; docs/architecture.md; docs/refactor-status.md.

Appendix: sources

Local evidence, relative to MetaG_new

- `artifacts/results/metag_opentecr_calibrated/`
- `artifacts/benchmark/comparison/`
- `artifacts/benchmark/paper_reproduction/`
- `artifacts/benchmark/curated/reference_report.json`
- `analysis/code_review/modelseed_preflight.json`
- `analysis/state_qc_20261007/`

External

Fan et al., dGbyG, Cell Systems (2025), DOI: 10.1016/j.cels.2025.101393.
OMol25: quantum-chemical training data and model scope.
Beber et al., eQuilibrator 3.0.

Evidence: Figures are PNG-only. Deck source: slides/MetaG.tex; figure generator: slides/make_figures.py.